

Your Students Don't Use LLMs Like You Wish They Did

你的学生并没有像你希望的那样使用 LLMs

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Abstract

摘要

Educational NLP systems are typically evaluated using engagement metrics and satisfaction surveys, which are at best a proxy for meeting pedagogical goals. We introduce six computational metrics for automated evaluation of pedagogical alignment in student-AI dialogue. We validate our metrics through analysis of 12,650 messages across 500 conversations from four courses. Using our metrics, we identify a fundamental misalignment: educators design conversational tutors for sustained learning dialogue, but students mainly use them for answer-extraction. Deployment context is the strongest predictor of usage patterns, outweighing student preference or system design: when AI tools are optional, usage concentrates around deadlines; when integrated into course structure, students ask for solutions to verbatim assignment questions. Whole-dialogue evaluation misses these turn-by-turn patterns. Our metrics will enable researchers building educational dialogue systems to measure whether they are achieving their pedagogical goals.

教育类自然语言处理系统通常使用参与度指标和满意度调查来评估，而这些指标最多只能作为实现教学目标的替代指标。我们引入了六个计算指标，用于自动评估学生与人工智能对话中的教学一致性。我们通过分析四个课程中 500 次对话的 12,650 条消息，验证了这些指标。使用我们的指标，我们发现了一个根本性的不一致：教育者设计对话式辅导系统以实现持续学习的对话，但学生主要使用它们来提取答案。部署环境是使用模式的最强预测因素，其影响超过了学生偏好或系统设计：当人工智能工具是可选时，使用集中在截止日期附近；当整合到课程结构中时，学生会询问与作业问题完全一致的解答。整体对话评估忽略了这些逐轮模式。我们的指标将使构建教育对话系统的研究人员能够衡量他们是否实现了教学目标。

1 Introduction

1 引言

Looking at 10 recent ACL papers that tested LLM educational systems with real students, 9 relied on satisfaction surveys and self-reported learning gains, without verifying whether students actually used the system as intended. This mismatch is an issue because extensive educational psychology research has shown that students are systematically poor judges of their own learning, suffering from metacognitive biases like the “illusion of fluency” that cause them to mistake ease of interaction for depth of understanding (Koriat and Bjork, 2005).

查看 10 篇最近的 ACL 论文，这些论文测试了使用真实学生进行 LLM 教育系统的实验，其中 9 篇依赖于满意度调查和自我报告的学习成果，而没有验证学生是否按照预期使用了该系统。这种不匹配是一个问题，因为大量的教育心理学研究已经表明，学生在判断自己的学习效果时系统性地存在不足，他们受到诸如“流畅性错觉”等元认知偏见的影响，会将交互的简便性误认为理解的深度（Koriat 和 Bjork, 2005）。

This paper introduces a computational framework for measuring what actually happens when students interact with educational AI systems. We introduce six metrics that capture distinct dimensions of behaviour: conversational engagement, learning orientation, scaffolding resistance, assignment dependency, crisis-mode behaviour, and usage concentration. These metrics provide a novel evaluation infrastructure for validating and analysing educational systems against their intended pedagogical function.

本文介绍了一个计算框架，用于衡量学生与教育类人工智能系统互动时实际发生的情况。我们引入了六个指标，以捕捉行为的不同维度：对话参与度、学习导向、支架抵抗性、作业依赖性、危机模式行为和使用集中度。这些指标为验证和分析教育系统是否符合其预期的教学功能提供了全新的评估基础设施。

Through analysis of 12,650 messages across 500 student-AI conversations from four courses, we uncover a fundamental misalignment between pedagogical design and student usage patterns. When AI tools are optional, usage concentrates around exam deadlines as crisis management, with 59% of all semester interactions occurring during a single exam week. When integrated into course structure, students frequently paste assignment questions directly to obtain solutions. These answer-seeking behaviours are masked in whole-dialogue evaluation, but clear in our turn-by-turn analysis. Deployment context shows stronger association with these problematic usage patterns than student preference or system design.

通过对来自四门课程的 500 个学生-人工智能对话中的 12,650 条消息进行分析，我们发现教学设计与学生使用模式之间存在根本性的不匹配。当 AI 工具是可选时，使用集中在考试截止日期附近，作为危机管理，所有学期互动中有 59% 发生在单个考试周内。当 AI 工具被整合到课程结构中时，学生经常直接粘贴作业问题以获取解答。这些寻求答案的行为在整体对话评估中被掩盖，但在我们的逐回合分析中却非常明确。部署背景显示出与这些有问题的使用模式更强的关联，而不仅仅是学生偏好或系统设计。

Our framework brings established educational data mining insights into computational dialogue assessment, enabling NLP researchers to move beyond satisfaction surveys toward evidence-based evaluation. We demonstrate that our metrics expose pedagogical failures invisible to standard engagement and satisfaction measures. Crucially, while student preferences for efficiency remain constant, deployment context determines whether these preferences lead to productive engagement or system gaming. External validation on RECIPE4U, a dataset with per-response satisfaction ratings, confirms that satisfaction does not correlate with pedagogical alignment. Our metrics will enable NLP researchers to develop and analyse systems that are truly educational, not just fun to engage with.

我们的框架将已有的教育数据挖掘见解引入计算对话评估领域，使自然语言处理（NLP）研究人员能够超越满意度调查，转向基于证据的评估。我们证明了我们的指标能够揭示标准参与度和满意度测量所无法察觉的教学失败。关键的是，尽管学生对效率的偏好保持不变，但部署环境决定了这些偏好是导致有效参与还是系统作弊。在 RECIPE4U 数据集上的外部验证表明，满意度与教学一致性之间没有相关性。我们的指标将使 NLP 研究人员能够开发和分析真正具有教育意义的系统，而不仅仅是令人愉悦的参与工具。

2 Related Work

2 相关工作

Three key issues create the conditions for pedagogical misalignment: (1) well-documented but ignored patterns of problematic student behaviour, (2) metacognitive limitations that make

三个关键问题造成了教学错位的条件：(1) 有充分记录但被忽视的不良学生行为模式，(2) 导致元认知能力受限的限制，使

self-assessment unreliable, and (3) flaws in how educational NLP systems are evaluated.

自我评估不可靠，以及 (3) 教育自然语言处理系统评估方法中存在的缺陷。

2.1 Documented but Ignored Student Behaviours

2.1 文档记录但被忽视的学生行为

Systems assume dialogue, students seek answers. Even at the system level, maintaining Socratic dialogue proves difficult: Ashok Kumar and Lan (2024) find that LLMs tasked with generating Socratic questions frequently produce outputs that directly reveal solutions, yet evaluate their fix using automatic generation metrics rather than verifying pedagogical outcomes with real students. This gap between intended pedagogy and actual function persists at the deployment level, where usage prioritises answer-seeking over exploratory learning.

系统假设对话，学生寻求答案。即使在系统层面，维持苏格拉底式对话也证明是困难的：Ashok Kumar 和 Lan (2024) 发现，被要求生成苏格拉底问题的大型语言模型（LLMs）经常产生直接揭示答案的输出，却使用自动生成指标来评估其效果，而不是通过真实学生验证教学成果。这种预期教学与实际功能之间的差距在部署层面依然存在，其中使用优先考虑寻求答案而非探索性学习。

Controlled studies miss crisis behaviour. Researchers typically recruit motivated volunteers rather than exhausted students facing deadlines. This selection bias artificially validates pedagogical assumptions and misses crisis-driven usage patterns. While Xu et al. (2024) note that model efficacy “diminishes with more complex teaching practices” in real classrooms, no prior work has examined how deployment strategies affect usage patterns.

受控研究忽略了危机行为。研究人员通常招募有动力的志愿者，而不是面临截止日期的疲惫学生。这种选择偏差人为地验证了教育假设，并忽略了由危机驱动的使用模式。尽管徐等人（2024）指出，在真实课堂中，模型效果“随着更复杂的教学实践而减弱”，但此前没有研究探讨部署策略如何影响使用模式。

Gaming behaviours are well-documented but rarely measured in NLP systems. Learning analytics research has extensively documented the behaviours we observe. Vanacore et al. (2024) found that 94.18% of students engage in “gaming the system” behaviours, with Baker et al. (2008) establishing a framework for understanding gaming as exploiting properties of the system rather than learning. Chiang et al. (2024) document students “manipulating the LLM to output specific strings” for high scores without meeting rubrics, yet such gaming behaviours are rarely measured.

游戏行为在文献中已有大量记录，但在自然语言处理（NLP）系统中却很少被测量。学习分析研究已经广泛记录了我们所观察到的行为。Vanacore 等人（2024）发现，94.18% 的学生参与了“利用系统”（gaming the system）的行为，Baker 等人（2008）则建立了一个理解游戏行为的框架，即把游戏视为利用系统特性而非学习。Chiang 等人（2024）记录了学生“操纵大型语言模型（LLM）以输出特定字符串”来获得高分，而无需满足评分标准，然而此类游戏行为却很少被测量。

Students concentrate usage around deadlines. Yang et al. (2020) identified procrastination patterns with submission rates accelerating near deadlines, while Ferguson and Clow (2016) found that MOOC engagement patterns consistently diverge from designers’ expectations. These studies explain why optional AI tools become crisis management systems.

学生通常在截止日期前集中使用学习工具。Yang 等人（2020）发现，拖延行为在接近截止日期时提交率会加速，而 Ferguson 和 Clow（2016）则发现，MOOC 的学习参与模式始终与设计者的预期存在差异。这些研究解释了为什么可选的 AI 工具会成为危机管理系统。

Scaffolding resistance reflects unproductive help-seeking. Aleven et al. (2004) found that 72% of help-seeking in intelligent tutoring systems represents unproductive gaming, consistent with their earlier findings (Aleven and Koedinger, 2001). Murray and VanLehn (2005) demonstrated that minimal friction significantly reduces unproductive help-seeking, supporting our finding that deployment context shapes behaviour more than student preferences.

脚手架阻力反映了非生产性的求助行为。Aleven 等人（2004）发现，智能教学系统中 72% 的求助行为属于非生产性的游戏行为，这与他们之前的发现（Aleven 和 Koedinger, 2001）一致。Murray 和 VanLehn（2005）证明，最小的摩擦力显著减少了非生产性的求助行为，支持了我们的发现，即部署环境比学生偏

好更能塑造行为。

NLP evaluation ignores established evidence of student gaming. While the gaming behaviours and temporal patterns above come from educational data mining, they have not been systematically incorporated into NLP evaluation of educational dialogue systems. Similarly, HCI research has examined student-AI interaction, revealing over-reliance and direct question copy-pasting in LLM-based assistants (Kazemitabaar et al., 2024), and exploring conversational agents for educational feedback (Wambsganss et al., 2020). However, this prior work focuses on system design and user studies rather than computational metrics for detecting pedagogically misaligned behaviours at scale. Our framework addresses this gap.

NLP 评估忽略了学生游戏行为的已有证据。尽管上述游戏行为和时间模式来自教育数据挖掘，但它们尚未被系统地纳入教育对话系统的 NLP 评估中。同样，人机交互（HCI）研究已考察了学生与 AI 的互动，揭示了基于大型语言模型（LLM）的助手存在过度依赖和直接复制粘贴问题（Kazemitabaar 等，2024），并探讨了用于教育反馈的对话代理（Wambsganss 等，2020）。然而，这些先前的研究更侧重于系统设计和用户研究，而不是大规模检测教学不一致行为的计算指标。我们的框架解决了这一差距。

2.2 The Metacognition Crisis

2.2 元认知危机

The “illusion of fluency” is just one aspect of systematic metacognitive failures that undermine self-directed learning with AI. The Dunning-Kruger effect manifests strongly in educational contexts. Kruger and Dunning (1999) found bottom-quartile students estimated their performance at the 62nd percentile, a 50-point overestimation. This “dual burden” means students least capable of learning effectively are most confident in their approaches.

“流畅感的错觉”只是系统性元认知失败的一个方面，这些失败削弱了自主学习 AI 的能力。杜宁-克鲁格效应在教育情境中表现得尤为明显。克鲁格和杜宁（1999）发现，排名后 25% 的学生估计自己的表现处于第 62 百分位，这相当于高估了 50 个百分点。这种“双重负担”意味着学习能力最差的学生，往往对自己的学习方法最为自信。

Students systematically misjudge effective learning. Bjork and Bjork (2011) demonstrate that students confuse retrieval strength with storage strength, choosing comfortable but ineffective methods. Koriat and Bjork (2005) found strong overconfidence effects when material feels fluent. These findings, including consistently poor self-assessment accuracy (Händel and Fritzsche, 2016; Metcalfe, 2009), explain why features that increase satisfaction ratings may undermine learning, since fluency feels like understanding and students reward the very features that reduce productive struggle.

学生系统性地错误判断有效的学习方式。Bjork 和 Bjork（2011）表明，学生会将检索强度与存储强度混淆，选择舒适但无效的学习方法。Koriat 和 Bjork（2005）发现，当材料感觉流畅时，会出现强烈的过度自信效应。这些研究结果，包括持续较低的自我评估准确性（Händel 和 Fritzsche, 2016; Metcalfe, 2009），解释了为什么那些提高满意度特征可能会削弱学习，因为流畅感让人感觉像是理解了，而学生则奖励那些减少有效努力的特征。

This creates the “satisfaction-effectiveness inversion”: features that make students rate AI tutors highly may undermine pedagogical value. Smooth conversational flow prevents productive struggle. Immediate answers short-circuit reflection. High availability enables procrastination rather than sustained engagement.

这产生了“满意度-有效性倒置”现象：那些使学生高度评价 AI 导师的功能可能会削弱教育价值。流畅的对话流程会阻碍有效的学习挣扎。即时回答会阻碍反思。高可用性反而助长拖延，而非持续参与。

2.3 Emerging Recognition of Evaluation Failures

2.3 评估失败的逐渐被认识

Technical metrics mask pedagogical failure. Standard NLP evaluation metrics fail to capture pedagogical effectiveness (Tack et al., 2023). Sonkar et al. (2024) acknowledge that supervised fine-tuning “doesn’t explicitly favor pedagogically effective responses,” highlighting challenges in aligning systems with pedagogical goals.

技术指标掩盖了教学上的失败。标准的自然语言处理评估指标未能捕捉到教学效果（Tack 等，2023）。Sonkar 等（2024）承认，有监督的微调“并未明确偏向于教学有效的响应”，突显了将系统与教学目标对齐的挑战。

High-performing learner bias reinforces gaps. Gurin Schleifer et al. (2024) document that LLM embeddings can identify correct student responses but cannot distinguish between different error patterns, meaning systems designed for equitable support fail to detect and help struggling learners, actually reinforcing achievement gaps. While LLMs will improve at detecting errors (a technical limitation), the fundamental bias persists: students who most need help are least equipped to extract it from conversational interfaces (Aleven and Koedinger, 2000), a structural issue in how these systems reward academic communication skills.

高表现学习者偏差会加剧差距。Gurin Schleifer 等人（2024）指出，LLM 嵌入可以识别正确的学生回答，但无法区分不同的错误模式，这意味着旨在提供公平支持的系统无法检测并帮助有困难的学习者，实际上加剧了成绩差距。虽然 LLM 在检测错误方面会有所改进（这是一个技术限制），但根本性的偏差依然存在：最需要帮助的学生往往最不擅长从对话界面中提取帮助（Aleven 和 Koedinger, 2000），这是这些系统在奖励学术交流技能方面存在的一种结构性问题。

Recent work acknowledges the evaluation gaps our framework addresses. Our emphasis on multidimensional metrics aligns with Rachatasumrit et al. (2024), who call for interpretable, meaningful metrics over prediction accuracy. Zambrano et al. (2024) support our use of multiple complementary metrics by demonstrating that self-report and observational measures produce substantially different results. This disconnect reflects broader patterns in educational technology where system features designed to help can inadvertently harm learning. The ‘assistance dilemma’ described by (Koedinger and Aleven, 2007) demonstrates that too much help prevents productive struggle, while too little causes frustration. This finding directly parallels our discovery that answer-seeking interactions, while satisfying to students, undermine pedagogical goals.

近期的研究承认了我们框架所解决的评估差距。我们对多维指标的重视与 Rachatasumrit 等人（2024）的观点一致，他们呼吁使用可解释且有意义的指标，而非仅仅关注预测准确性。Zambrano 等人（2024）通过证明自我报告和观察性测量会产生显著不同的结果，支持了我们使用多种互补指标的做法。这种脱节反映了教育技术领域更广泛的模式，即旨在帮助学习者的系统功能可能无意中阻碍学习。Koedinger 和 Aleven（2007）描述的“帮助困境”表明，过多的帮助会妨碍有成效的探索，而过少的帮助则会导致挫败感。这一发现与我们发现的结论直接呼应：虽然寻求答案的互动对学生来说令人满意，但会削弱教学目标。

Further, (Vanzo et al., 2025) celebrate GPT-4 as a homework tutor based on student satisfaction and engagement metrics, with 91% of students wanting to continue using the system. However, their own analysis reveals that engagement, not the treatment condition, predicted learning gains, and improvements were limited to objective-type exercises where answer extraction would be most effective. Critically, their engagement metrics (word count and message frequency) cannot distinguish between productive learning dialogue and answer extraction behaviours, a distinction our framework makes through turn-by-turn analysis and multiple convergent metrics. Without such granular analysis, what appears as beneficial engagement may actually represent the very gaming behaviours that undermine learning. Berman et al. (2024) explicitly distinguish between usability and effectiveness evaluation, noting most AI tools are assessed on ease of use rather than goal achievement, a distinction central to our framework.

此外，(Vanzo 等，2025) 根据学生满意度和参与度指标，称赞 GPT-4 作为家庭作业辅导工具，有 91% 的学生希望继续使用该系统。然而，他们自己的分析表明，参与度而非干预条件预测了学习成果的提升，

而改进仅限于答案提取效果最明显的客观型练习题。关键的是，他们的参与度指标（字数和消息频率）无法区分有益的学习对话和答案提取行为，而我们的框架通过逐轮分析和多种聚合指标来做出这种区分。如果没有这种细致的分析，表面上看似有益的参与度可能实际上代表了削弱学习的作弊行为。Berman 等 (2024) 明确区分了可用性与有效性评估，指出大多数人工智能工具是基于易用性而非目标达成来进行评估的，这一点对于我们的框架至关重要。

Recent work has proposed complementary evaluation approaches. Maurya et al. (2025) introduce a taxonomy of eight pedagogical dimensions for assessing AI tutor response quality, while Oliveira et al. (2025) propose the DRIVE framework for evaluating student learning through GenAI interactions, finding that assessment design shapes whether students use AI for idea development or passive task delegation. Both share our concern with moving beyond satisfaction-based evaluation but differ in focus: Maurya et al. (2025) evaluate the tutor's quality per-response, Oliveira et al. (2025) assess learning outcomes through interaction analysis. Our work provides a complementary layer: automated metrics for detecting student behavioural patterns such as crisis-driven usage and scaffolding resistance. These patterns emerge only when analysing sequences of interactions over time, making them invisible to per-response evaluation and difficult to capture through outcome-focused assessment alone.

近期的研究提出了互补的评估方法。Maurya 等人 (2025) 引入了一个包含八个教学维度的分类法，用于评估 AI 教学助手的响应质量，而 Oliveira 等人 (2025) 则提出了 DRIVE 框架，用于通过生成式人工智能 (GenAI) 互动评估学生的学习，发现评估设计决定了学生是使用 AI 进行主动的创意发展还是被动的任务委托。两者都关注超越基于满意度的评估，但在侧重点上有所不同：Maurya 等人 (2025) 评估教学助手的每条响应质量，Oliveira 等人 (2025) 则通过互动分析评估学习成果。我们的工作提供了一个互补的层面：自动指标用于检测学生的行为模式，例如危机驱动的使用和支架抵抗。这些模式只有在分析随时间推移的一系列互动时才会显现出来，因此无法被每条响应的评估所察觉，也难以通过以结果为导向的评估单独捕捉。

3 Behaviour Metrics

3 行为指标

We developed a computational framework to detect and quantify student usage behaviours in educational AI interactions. Our approach uses six novel metrics, validated against human expert judgement (Section 4.2), enabling scalable detection of usage patterns that diverge from pedagogical intent.

我们开发了一个计算框架，用于检测和量化学生在教育人工智能交互中的使用行为。我们的方法采用了六个新颖的指标，并经过人类专家判断的验证（第 4.2 节），从而能够可扩展地检测偏离教学意图的使用模式。

3.1 Metrics Framework

3.1 指标框架

We introduce six metrics designed to capture behaviours relevant to pedagogical alignment in student-AI dialogue. To our knowledge, these are the first computational metrics specifically designed for evaluation of such behaviours at scale. We implemented two analysis approaches: turn-by-turn analysis evaluates each student-AI exchange independently, capturing granular behavioural patterns like immediate answer-seeking and scaffolding resistance; whole dialogue anal-

我们介绍了六个指标，旨在捕捉与教学一致性相关的行为，在学生与 AI 的对话中。据我们所知，这些是首个专门设计用于大规模评估此类行为的计算指标。我们实施了两种分析方法：回合分析 (turn-by-turn analysis) 独立评估每个学生与 AI 的交流，捕捉诸如立即寻求答案和抗拒支架式教学等细致的行为模式；整体对话分析 (whole dialogue analysis) 则...

ysis processes entire conversations as single units, identifying broader patterns like overall engagement trajectory and assignment characteristics.

分析过程将整个对话视为单一单位，识别更广泛的模式，如整体参与度趋势和任务特征。

Four metrics employ large language models for analysis: LOI, SRS, ADR, and CMI. These LLM-based metrics use zero-shot prompting to measure behavioural indicators, with complete prompts provided in Appendix A. All metric weights were set based on the authors' teaching experience and pedagogical judgment, without data-driven tuning. For all metrics, the LLM assigns continuous 0–1 scores rather than binary classifications, allowing proportional values for ambiguous cases (see Appendix A.3 for examples).

四种指标利用大型语言模型进行分析：LOI、SRS、ADR 和 CMI。这些基于 LLM 的指标使用零次提示 (zero-shot prompting) 来衡量行为指标，完整的提示内容见附录 A。所有指标的权重均根据作者的教学经验和教育判断进行设置，而非基于数据驱动的调优。对于所有指标，LLM 会分配连续的 0–1 分数，而不是二元分类，从而允许在模糊情况下使用比例值（有关示例，请参见附录 A.3）。

3.1.1 Conversational Engagement Score (CES)

3.1.1 对话参与度评分 (CES)

Drawing on dialogue analysis literature on productive educational discourse (Chi and Wylie, 2014), CES distinguishes genuine dialogue from transactional information-seeking through a weighted average of four dimensions: log-normalised turn count (sustained interaction), follow-up rate (responses that build on AI answers rather than starting new topics), context reference rate (callbacks to earlier discussion), and acknowledgement rate (engagement markers like “I see” or “that makes sense”). For LLM-analysed components (follow-up, context reference, acknowledgement), we apply binary classification to each student turn, with the final rate calculated as the fraction of messages labeled as positive. Values range 0–1, with higher scores indicating more conversational engagement. See Appendix A.2.1 for implementation details.

借鉴关于富有成效的教育对话的对话分析文献 (Chi 和 Wylie, 2014)，CES 通过四个维度的加权平均值来区分真正的对话与交易性信息获取：对数归一化的发言次数（持续互动）、后续率（基于 AI 回答进行回应而非开启新话题的回应）、上下文引用率（回调先前讨论）以及确认率（如“我明白了”或“这有道理”等参与性标记）。对于由大语言模型分析的组成部分（后续率、上下文引用率、确认率），我们对每个学生的发言进行二元分类，最终的比率计算为标记为正面的讯息比例。数值范围为 0–1，得分越高表示对话参与度越高。详见附录 A.2.1 的实现细节。

3.1.2 Learning Orientation Index (LOI)

3.1.2 学习取向指数 (LOI)

Adapting Bloom's Taxonomy (Bloom et al., 1956) and help-seeking research (Aleven et al., 2004), LOI measures the balance between exploratory learning and solution-seeking through LLM classification. Each student message receives a score (0.0–1.0) distinguishing exploratory markers (process-focused questions, conceptual connections, hypothetical scenarios) from solution-seeking markers (direct answer requests, template seeking, results-focused queries). The index represents the proportion of exploratory interactions:

借鉴布卢姆分类法 (Bloom 等人, 1956) 和求助行为研究 (Aleven 等人, 2004)，LOI 通过 LLM 分类衡量探索性学习与求解行为之间的平衡。每条学生消息都会获得一个分数 (0.0–1.0)，以区分探索性标记（过程性问题、概念联系、假设情景）与求解性标记（直接答案请求、模板寻求、结果导向的查询）。该指数代表了探索性互动的比例：

$$LOI = \frac{\sum_i \text{exploratory_weight}_i}{\sum_i \text{all_weights}_i}$$

Values approaching 1 indicate learning-oriented behaviour; values near 0 indicate answer-extraction patterns¹. See Appendix A.2.2 for classification criteria.

值接近 1 表示学习导向的行为；值接近 0 表示答案提取模式¹。参见附录 A.2.2 中的分类标准。

3.1.3 Scaffolding Resistance Score (SRS)

3.1.3 支架阻力评分 (SRS)

Building on gaming detection research (Baker et al., 2008), SRS quantifies student rejection of pedagogical guidance through a three-step process. First, it identifies AI scaffolding attempts (hints, leading questions, Socratic method). Second, it classifies student responses as accepting (engages with guidance), resisting (requests direct answer), or bypassing (ignores guidance). Finally, it calculates resistance proportion:

基于游戏检测研究 (Baker 等, 2008 年), SRS 通过一个三步流程量化学生对教学指导的拒绝程度。首先, 它识别人工智能支架尝试 (提示、引导性问题、苏格拉底方法)。其次, 它将学生回答分类为接受 (参与指导)、抵抗 (请求直接答案) 或绕过 (忽略指导)。最后, 它计算抵抗比例:

$$SRS = \frac{\text{resist_count} + 0.5 \times \text{bypass_count}}{\text{total_scaffolding_attempts}}$$

Higher scores indicate greater resistance to pedagogical scaffolding, suggesting preference for direct answers over guided learning. See Appendix A.2.3 for detection criteria.

更高的分数表明对教学支架具有更大的抵抗力, 这表明更倾向于直接答案而非指导性学习。参见附录 A.2.3 中的检测标准。

3.1.4 Assignment Dependency Ratio (ADR)

3.1.4 任务依赖比率 (ADR)

Combining structural text analysis with teaching observation of homework-driven query patterns, ADR detects assignment-driven usage through parallel rule-based and LLM-based analysis. Rule-based detection identifies structural markers (numbered questions, academic imperatives). LLM analysis evaluates conversational patterns (topic jumping, problem set behaviour). Both methods produce scores from 0–1:

结合结构化文本分析与作业驱动查询模式的教学观察, ADR 通过并行的基于规则和基于大语言模型 (LLM) 的分析, 检测作业驱动的使用情况。基于规则的检测识别结构化标记 (如编号问题、学术性命令)。基于 LLM 的分析评估对话模式 (如话题跳跃、问题集行为)。两种方法均产生 0–1 分的评分:

$$ADR_{\text{rule}} = \frac{\text{detected_markers}}{\text{total_messages}}$$

$$ADR_{\text{llm}} = \text{assignment_probability}$$

Higher values indicate greater likelihood of assignment-related usage. See Appendix A.2.4 for implementation details.

较高值表示更有可能用于与分配相关的用途。有关实现细节，请参见附录 A.2.4。

3.1.5 Crisis Mode Indicator (CMI)

3.1.5 危机模式指示器（CMI）

Adapting procrastination research (Yang et al., 2020), CMI detects behavioural shifts during high-pressure periods through within-student temporal analysis. It compares peak usage against base-line behaviour using five weighted indicators: panic indicators (urgency language like “desperate” or “exam tomorrow”), query directness shift (more blunt requests), late-night usage (activity outside normal hours), single-exchange ratio (one-and-done queries), and engagement decrease (shorter

借鉴拖延研究（Yang 等，2020），CMI 通过学生内部的时间分析，检测在高压时期的行为变化。它使用五个加权指标，将峰值使用情况与基线行为进行比较：恐慌指标（如“绝望”或“考试明天”等表达紧迫性的语言）、查询直接性变化（更直接的请求）、深夜使用（在非正常时间段的活动）、单次交流比例（一次性问题）、参与度下降（更短的互动）。

	LOI			CES			SRS			ADR		
r	Exact	±1	r	Exact	±1	r	Exact	±1	r	Exact	±1	
GPT 4.1-mini Turn	0.62	66%	97%	0.42	50%	95%	0.64	56%	88%	-	-	-
GPT 4.1-mini Whole	0.33	16%	44%	0.21	10%	58%	0.25	39%	75%	0.22	48%	65%
GPT 5 Turn	0.72	72%	99%	0.59	44%	92%	0.67	63%	91%	-	-	-
GPT 5 Whole	0.47	19%	63%	0.46	34%	81%	0.49	47%	79%	0.31	38%	54%
Rule-based	-	-	-	-	-	-	-	-	-	0.35	82%	85%
Human-Human†	0.58	59%	100%	0.67	55%	88%	0.64	60%	85%	0.65	75%	82%

	LOI			CES			SRS			ADR		
r	精确	±1	r	精确	±1	r	精确	±1	r	精确	±1	
GPT 4.1-mini Turn	0.62	66%	97%	0.42	50%	95%	0.64	56%	88%	-	-	-
GPT 4.1-mini 整体	0.33	16%	44%	0.21	10%	58%	0.25	39%	75%	0.22	48%	65%
GPT 5 发布	0.72	72%	99%	0.59	44%	92%	0.67	63%	91%	-	-	-
GPT 5 全部	0.47	19%	63%	0.46	34%	81%	0.49	47%	79%	0.31	38%	54%
规则驱动	-	-	-	-	-	-	-	-	-	0.35	82%	85%
人-人 †	0.58	59%	100%	0.67	55%	88%	0.64	60%	85%	0.65	75%	82%

Table 1: Model-human and human-human agreement. r = Pearson correlation (top 5 rows), and weighted Cohen’s kappa (bottom row). ±1 = off-by-one accuracy. † row is based on 100 conversations, while the others are based on 248. GPT-5 turn-by-turn achieves highest correlations and consistently outperforms whole-dialogue analysis.

表 1：模型与人类、人类与人类之间的共识。r = 皮尔逊相关系数（前 5 行），加权 Cohen’s kappa（最后一行）。±1 = 偏差为 1 的准确率。† 行基于 100 次对话，而其他行基于 248 次对话。GPT-5 的逐回合分析实现了最高的相关性，并且始终优于整体对话分析。

responses, fewer follow-ups). Values range 0–1, with higher scores indicating crisis-driven usage patterns. See Appendix A.2.5 for baseline establishment and shift detection methods.

响应次数较少。评分范围为 0–1，分数越高，表示使用模式越倾向于危机驱动。详见附录 A.2.5 中的基线建立和变化检测方法。

3.1.6 Usage Concentration Index (UCI)

3.1.6 使用集中度指数 (UCI)

Applying the Gini coefficient from economics to temporal usage patterns, UCI measures the distribution of platform usage across a semester through three components: Gini coefficient (inequality in daily usage distribution), normalised peak-to-average ratio (intensity of usage spikes), and temporal clustering (consecutive high-usage periods). Values approaching 1 indicate highly concentrated crisis-driven usage; lower values suggest distributed engagement throughout the semester. See Appendix A.2.6 for complete formulas.

将经济学中的基尼系数应用于时间使用模式，UCI 通过三个组成部分衡量平台使用在学期内的分布情况：基尼系数（日常使用分布的不平等程度）、归一化峰值与平均值比（使用高峰的强度）以及时间聚类（连续的高使用时段）。数值接近 1 表示高度集中于危机驱动的使用；较低的数值则表明在整个学期中分布较为广泛。完整公式请参见附录 A.2.6。

3.2 Design Rationale

3.2 设计理由

Our metrics rely on heuristic weighting and zero-shot LLM classification. We explain three key design choices below.

我们的指标依赖于启发式加权和零样本 LLM 分类。我们在下面解释三个关键的设计选择。

Weighting. All component weights reflect pedagogical priorities rather than data-driven optimisation, as no ground-truth pedagogical outcome data exists against which to tune. Within each metric, the highest-weighted components are those most directly observable and discriminating. For CES, turn count (0.40) receives the highest weight as the strongest signal distinguishing dialogue from transaction, followed by follow-up rate (0.25), context reference (0.20), and acknowledgement (0.15). For CMI, panic indicators (0.30) and query directness shift (0.25) are weighted highest as the most direct behavioural signals of crisis-mode usage, with late-night usage (0.20), single-exchange ratio (0.15), and engagement decrease (0.10) as corroborating indicators.

加权。所有组件权重反映的是教育优先级，而非数据驱动的优化，因为不存在可用于调整的真实教育结果数据。在每个指标中，权重最高的组件是最直接可观察且具有区分性的。对于 CES（对话性评分），对话轮数（0.40）具有最高权重，因为它是区分对话与交易的最强信号，其次是跟进率（0.25）、上下文引用（0.20）和确认（0.15）。对于 CMI（危机模式使用指标），恐慌指标（0.30）和查询直接性变化（0.25）具有最高权重，因为它们是最直接的行为信号，表明用户处于危机模式，随后是深夜使用（0.20）、单次交流比率（0.15）和参与度下降（0.10）作为辅助指标。

For ADR, copy-paste indicators (0.40) receive the highest weight as the strongest signal of assignment dependency, followed by problem set behaviour (0.30), answer-seeking focus (0.20), and urgency signals (0.10). For SRS, response weights follow a graduated scale: resisting (1.0) represents clear rejection of scaffolding, bypassing (0.5) reflects passive avoidance, and mixed (0.25) indicates partial engagement.

对于 ADR，复制粘贴指标（0.40）权重最高，是任务依赖性的最强信号，其次是问题集行为（0.30）、寻求答案的专注度（0.20）以及紧迫性信号（0.10）。对于 SRS，响应权重遵循一个渐进的等级：抵抗（1.0）表

示明确拒绝支架，绕过 (0.5) 反映了被动回避，混合 (0.25) 表示部分参与。

Continuous scoring. We chose continuous 0–1 scores over binary classification because pedagogical behaviours exist on a spectrum. A student may shift from exploratory to answer-seeking within a single conversation, or partially engage with scaffolding before requesting a direct answer. Binary labels would obscure these within-conversation dynamics that are central to our analysis. Appendix Table 4 illustrates how continuous scoring captures these ambiguous cases.

连续评分。我们选择了连续的 0–1 评分而非二元分类，因为教学行为存在于一个连续谱上。一名学生可能在一次对话中从探索性行为转变为寻求答案的行为，或者在请求直接答案之前部分参与支架式教学。二元标签会掩盖这些对话内的动态变化，而这些变化正是我们分析的核心。附录表 4 展示了连续评分如何捕捉这些模糊情况。

Zero-shot classification. We use zero-shot prompting without fine-tuning or few-shot examples for all LLM-based metrics. This choice prioritises generalisability across educational domains: fine-tuned classifiers would require labelled training data for each new deployment context, undermining the framework’s utility as a general evaluation tool. Our validation (Table 1) demonstrates that zero-shot GPT-5 turn-by-turn analysis achieves correlations of 0.59–0.72 with human judgement, approaching inter-rater agreement levels (0.58–0.67 weighted kappa), suggesting that the classification criteria in our prompts are sufficiently well-specified for reliable automated evaluation.

零样本分类。我们对所有基于大语言模型 (LLM) 的指标使用零样本提示，而不进行微调或使用少量样本示例。这一选择优先考虑了在教育领域中的泛化能力：经过微调的分类器需要为每个新的部署环境提供标记的训练数据，这会削弱该框架作为通用评估工具的实用性。我们的验证 (表 1) 表明，零样本 GPT-5 逐回合分析与人类判断之间的相关性达到 0.59–0.72，接近评分者间一致性水平 (0.58–0.67 加权 Kappa 值)，这表明我们的提示中的分类标准足以实现可靠自动化评估。

4 Experiments

4 实验

4.1 Dataset Composition

4.1 数据集组成

We analysed 500 student-AI conversations from five distinct educational datasets spanning three

我们分析了来自三个不同教育数据集的 500 个学生与 AI 的对话，这些数据集涵盖了五个不同的教育领域。

disciplines and two interaction paradigms (see Appendix A.4 for sampling strategy).

学科和两种交互范式 (参见附录 A.4 中的抽样策略)。

Pedagogical Support Tool Datasets: We have data from three courses where students interacted with optional AI tools designed to scaffold learning through guided questioning and progressive hints: DrMattTabolism, deployed in Week 3 of a second-year biochemistry course covering metabolism modules (Weeks 1-6); DrNucleicAlice, deployed later in the same course covering molecular biology modules (Weeks 7-12)²; MEDS2004, an optional tool in a second-year medical sciences course that provided choreographed practice with past exam questions, following a strict question-answer-feedback sequence; and OLiMent (Yuan et al., 2025), used in an introductory data science course for self-reflection on progress.

教学支持工具数据集：我们有三门课程的数据，这些课程中学生使用了可选的 AI 工具，这些工具通过引导式提问和逐步提示来支持学习：DrMattTabolism，用于第二年生化课程的代谢模块（第 1-6 周），在课程第 3 周部署；DrNucleicAlice，用于同一课程的分子生物学模块（第 7-12 周）²；MEDS2004，是第二

年医学科学课程中的一个可选工具，提供有计划的练习，使用过去的考试题目，遵循严格的问答反馈流程；以及 OLiMent (Yuan 等, 2025)，用于入门数据科学课程，用于自我反思学习进度。

While all tools were built on similar platforms with pedagogical constraints, their implementations varied. For example, DrMattTabolism employed a flexible questioning approach that would provide answers when pressed but consistently included prompting questions, resulting in shorter conversations (9.0 messages average); DrNucleicAlice enforced stricter Socratic dialogue requiring the AI to respond only with guiding questions until students demonstrated understanding, with scaffolding through Bloom's Taxonomy (Bloom et al., 1956), leading to longer interactions (14.5 messages average) as students were challenged to engage more deeply before receiving answers.

尽管所有工具都是在相似的平台构建，并受到教学限制，但它们的实现方式却各不相同。例如，DrMattTabolism 采用了一种灵活的提问方式，当被追问时会提供答案，但始终包含引导性问题，导致对话较短（平均 9.0 条消息）；而 DrNucleicAlice 则强制执行更严格的苏格拉底式对话，要求人工智能仅以引导性问题作答，直到学生表现出理解，通过布鲁姆分类法（Bloom 等, 1956）进行支架式教学，从而导致更长的互动（平均 14.5 条消息），因为学生在获得答案前被挑战着更深入地参与。

Unrestricted AI Dataset: The StudyChat (SC) dataset (McNichols et al., 2025) from an upper-level computer science course where students were encouraged to use an AI assistant throughout the semester for all coursework, represents a contrasting deployment paradigm where AI interaction was integrated into weekly assignments.

无限制 AI 数据集：来自一门高级计算机科学课程的 StudyChat (SC) 数据集（McNichols 等, 2025），该课程鼓励学生在整个学期中使用 AI 助手完成所有课程作业，代表了一种与之相对的部署范式，其中 AI 交互被整合到每周的作业中。

All datasets represent complete academic semesters (DrNucleicAlice was introduced slightly later in the semester) with student identifiers removed. The original data collection obtained consent from students for research purposes including dialogue analysis. Only StudyChat is publicly available; the remaining datasets cannot be released due to ethics restrictions and privacy requirements.

所有数据集均代表完整的学术学期（DrNucleicAlice 在学期中稍晚些时候才被引入），并且已去除学生标识符。原始数据收集过程中，已获得学生同意用于研究目的，包括对话分析。只有 StudyChat 数据集是公开的；其余数据集由于伦理限制和隐私要求而无法发布。

4.2 Model Configuration and Validation

4.2 模型配置与验证

Human Agreement To validate our metrics, we manually labeled 248 conversations (approximately 50 per dataset, with 2 held-out due to non-English text). The primary rater evaluated all 248 conversations while the second rater independently assessed 100 overlapping conversations to establish inter-rater reliability. The annotators bring complementary pedagogical backgrounds: one is an assistant professor in computer science with 4 years experience as the instructor for university courses; the other is a graduate student with four years of experience as a teaching assistant at university and secondary school levels. Human evaluators rated using a 1-5 scale for continuous metrics (CES, SRS, ADR) and using categories for LOI (Answer-seeking, Mixed, Exploratory). We found high inter-rater agreement (Table 1, bottom row): weighted Cohen's kappa (with quadratic weights) ranged from 0.58 for LOI to 0.67 for CES, with exact match rates of 55 – 75% and within-one-level agreement of 82 – 100%. These results confirm both the reliability of our annotation protocol and establish empirical baselines for evaluating model performance. We use all 248 labeled samples to evaluate the models.

人类共识为了验证我们的指标，我们手动标注了 248 个对话（每个数据集大约 50 个，其中 2 个由于文本非英语而保留下来）。主要标注者评估了所有 248 个对话，而第二个标注者独立评估了 100 个重叠的对话

以建立评分者间的一致性。标注者具有互补的教学背景：一个是拥有四年大学课程讲师经验的计算机科学助教；另一个是拥有四年大学和中学教学助理经验的研究生。人类评估者使用 1-5 分的连续指标（CES、SRS、ADR）进行评分，并使用类别对 LOI（寻求答案、混合、探索性）进行评分。我们发现评分者之间的一致性很高（表 1，最后一行）：使用二次权重的加权 Cohen’s kappa 值从 LOI 的 0.58 到 CES 的 0.67，精确匹配率是 55 — 75%，在同一级别内的同意率是 82 — 100%。这些结果确认了我们标注协议的可靠性，并为评估模型性能建立了实证基准。我们使用所有 248 个标注样本来评估模型。

LLM Selection We compared two models to evaluate metric reliability and cost-effectiveness. GPT-4.1-mini offered cost-effective processing whereas GPT-5 provided enhanced reasoning capabilities at higher cost with reasoning effort set at the default medium setting. All LLM-based metrics employed zero-shot prompting without examples or fine-tuning, with complete prompts provided in the respective appendix sections. Cost analysis is available in Appendix A.6.

LLM 选择 我们比较了两个模型以评估指标的可靠性和成本效益。GPT-4.1-mini 提供了成本效益高的处理方式，而 GPT-5 在较高的成本下提供了增强的推理能力，其推理努力设置为默认的中等水平。所有基于 LLM 的指标均采用零样本提示（没有示例或微调），完整的提示内容在相应的附录部分中提供。成本分析详见附录 A.6。

Model Validation ADR showed the weakest computational-human alignment. LLM-based approaches averaged 3.00 compared to human ratings of 1.49, an overestimation of +1.51 points. LLM ratings were within 1 point of human ratings only 55.1% of the time, with agreement of 0.33. The most significant miscalibration occurred in MEDS2004: computational methods flagged 72% of conversations as assignment-related while human evaluators identified only 7%. This demonstrates the challenge of reliably distinguishing homework copying from legitimate learning interactions through automated methods.

模型验证 ADR 显示了最弱的计算与人类一致性。基于 LLM 的方法平均得分为 3.00，而人类评分仅为 1.49，存在 +1.51 分的高估。LLM 评分与人类评分相差 1 分的情况仅占 55.1%，一致性为 0.33。最显著的校准错误出现在 MEDS2004 中：计算方法将 72% 的对话标记为作业相关，而人类评估者仅识别出 7%。这表明通过自动化方法可靠地区分抄袭作业与合法学习互动的挑战。

5 Results

5 结果

We applied our computational framework to detect and quantify distinct patterns of pedagogical misalignment across 500 student-AI conversations (temporal patterns are visualised in Appendix Figure 3). Table 2 shows the results of our metrics. For behavioural metrics (LOI, CES, SRS), we used GPT-5 with turn-by-turn analysis. For Assignment Dependency Ratio, we used GPT-5 with whole dialogue analysis, though as discussed in Section 6.1, computational detection significantly underperformed human evaluation.

我们将我们的计算框架应用于 500 次学生与 AI 对话中，以检测和量化教学不一致的不同模式（时间模式在附录图 3 中可视化）。表 2 展示了我们的指标结果。对于行为指标（LOI、CES、SRS），我们使用 GPT-5 进行逐回合分析。对于任务依赖度比率，我们使用 GPT-5 进行整体对话分析，尽管如第 6.1 节所述，计算检测在很大程度上不如人工评估的表现。

Learning Orientation Index (LOI) StudyChat, the unrestricted platform, showed substantially reduced learning-oriented behaviour. Only 2.0% of conversations were exploratory compared to 15.5% in constrained platforms (Table 3). There was more answer-seeking: 92.0% in StudyChat versus 66.5% in constrained platforms. When students have unrestricted access, they primarily seek direct answers rather than exploring concepts. Figure 2 in the appendices shows this inverse relationship between access and learning orientation.

学习导向指数 (LOI) 研究显示，StudyChat 这个无限制平台表现出显著减少的学习导向行为。仅有 2.0% 的对话是探索性的，而受限平台则达到 15.5%（见表 3）。此外，寻求答案的行为更多：StudyChat 中为 92.0%，而受限平台仅为 66.5%。当学生拥有无限制的访问权限时，他们主要寻求直接答案，而不是探索

概念。附录中的图 2 显示了访问权限与学习导向之间的反向关系。

Conversational Engagement Score (CES) The unrestricted platform achieved higher engagement metrics (0.713 vs 0.692), with 58% of conversations classified as highly conversational versus 47.5% for constrained platforms. However, these conversations correlated with lower learning orientation (see Appendix Figure 2). Higher engagement served answer extraction rather than understanding.

对话参与度评分（CES）无限制平台实现了更高的参与度指标（0.713 vs 0.692），其中 58% 的对话被归类为高度对话性，而受限平台为 47.5%。然而，这些对话与较低的学习导向相关（见附录图 2）。更高的参与度服务于答案提取，而非理解。

Scaffolding Resistance Score (SRS) Students systematically avoided pedagogical scaffolding: 49.0% ignored guidance, 20.8% reformulated queries to avoid it, and 18.3% directly requested answers. Resistance rates were similar between platforms (0.227 constrained, 0.223 unrestricted), indicating consistent avoidance regardless of system type. Students treat scaffolding as a barrier to obtaining answers.

支架阻力评分（SRS）学生系统性地回避了教学支架：49.0% 的学生忽视了指，20.8% 的学生重新表述问题以避免支架，18.3% 的学生直接请求答案。不同平台之间的阻力率相似（0.227 受限，0.223 无限制），表明无论系统类型如何，回避行为保持一致。学生将支架视为获取答案的障碍。

Assignment Dependency Ratio (ADR) Human evaluation revealed minimal assignment dependency across datasets: average rating of 1.49 on our 5-point scale, with 81% of conversations (198/245) rated as 1 (no dependency). Even MEDS2004, designed around past exam questions, showed only 7% of conversations with substantive assignment dependency (ratings above 1). This suggests students rarely copy-paste assignment text, instead framing homework questions conversationally.

作业依赖比率（ADR）人工评估显示，各数据集中的作业依赖程度非常低：我们在 5 分量表上的平均评分是 1.49，其中 81% 的对话（198/245）被评为 1（无依赖）。即使是围绕过去考试题目设计的 MEDS2004，也仅有 7% 的对话显示出实质性的作业依赖（评分高于 1）。这表明学生很少直接复制粘贴作业文本，而是以对话形式提出作业问题。

Dataset	LOI	CES	SRS	ADR		CMI	UCI
Rule	LLM						
DrMattTabolism	33	35	16	3	41	20	64
DrNucleicAlice	38	72	33	2	39	13	67
MEDS2004	13	67	27	2	72	14	74
OLiMent	34	70	16	5	26	18	67
StudyChat	15	71	22	12	58	-	39

数据集	LOI	消费电子展（Consumer Electronics Show）	SRS	ADR	CMI	UCI	
规则	大型语言模型						
DrMattTabolism	33	35	16	3	41	20	64
DrNucleicAlice	38	72	33	2	39	13	67
MEDS2004	13	67	27	2	72	14	74
OLiMent	34	70	16	5	26	18	67
StudyChat	15	71	22	12	58	-	39

Table 2: Summary of pedagogical misalignment metrics by dataset. All values are scaled 0-1

and shown as percentages (%). Higher values indicate: CES = conversational engagement, SRS = scaffolding avoidance, ADR = detection of assignment content, CMI/UCI = crisis-driven usage. For LOI, lower values indicate answer-seeking behaviour.

表 2：按数据集划分的教学不匹配指标摘要。所有值均按 0-1 进行标准化，并以百分比（%）显示。较高值表示：CES = 对话参与度，SRS = 避免支架使用，ADR = 检测作业内容，CMI/UCI = 危机驱动使用。对于 LOI，较低值表示寻求答案的行为。

Platform	AS	E	M
Constrained (n=400)	66.5	15.5	18.0
StudyChat (n=100)	92.0	2.0	6.0

平台	AS	E	M
受限 (n=400)	66.5	15.5	18.0
StudyChat (n=100)	92.0	2.0	6.0

Table 3: Learning Orientation Distribution by Platform Type. Answer Seeking, Exploratory and Mixed. All values shown are percentages (%).

表 3：按平台类型划分的学习取向分布。寻求答案、探索型和混合型。所列所有数值均为百分比（%）。

Crisis Mode Indicator (CMI) Students exhibited consistent behavioural shifts between baseline and assessment periods across optional pedagogical tools ³. Within-student comparisons revealed message lengths decreased 36-96% during assessments and panic indicators increased 1-19% from baseline levels. The overall CMI scores (0.13-0.20 across four datasets) quantify these within-student behavioural shifts, indicating students shift from exploratory learning during baseline periods to answer-extraction behaviours during assessments. See Appendix 4 for a detailed breakdown.

危机模式指标（CMI）学生在可选教学工具 ³ 的基线期和评估期间表现出一致的行为变化。对学生个体的比较显示，在评估期间信息长度减少了 36-96%，而恐慌指标从基线水平增加了 1-19%。总体 CMI 得分（四个数据集的 0.13-0.20）量化了这些个体行为变化，表明学生在基线期间从探索性学习转变为评估期间的答题提取行为。详见附录 4 中的详细分解。

5.1 Temporal Usage Patterns

5.1 时间使用模式

Pedagogically-Constrained Platforms The four constrained datasets revealed a consistent pattern: optional pedagogical tools function as crisis management systems rather than learning companions. With mean UCI of 0.681 (SD = 0.043), nearly half of all interactions compressed into single peak weeks. DrNucleicAlice showed this with 59.7% of total semester usage occurring during one exam week, while even the most distributed platform (OLiMent at 41.1%) still showed heavy crisis concentration. This pattern, consistent across different courses and instructors (Figure 1), suggests struc-

教学约束平台

这四个受限数据集揭示了一个一致的模式：可选的教学工具起着危机管理系统的作用，而不是学习伙伴。平均 UCI 为 0.681（标准差 = 0.043），几乎所有互动都集中在单个高峰期的周内。DrNucleicAlice 显示了这一点，该平台有 59.7% 的整个学期使用量集中在一周考试期间，即使是最分散的平台（OLiMent 在 41.1%）仍然显示出严重的危机集中现象。这种模式在不同课程和教师之间保持一致（图 1），表明结构上的.....

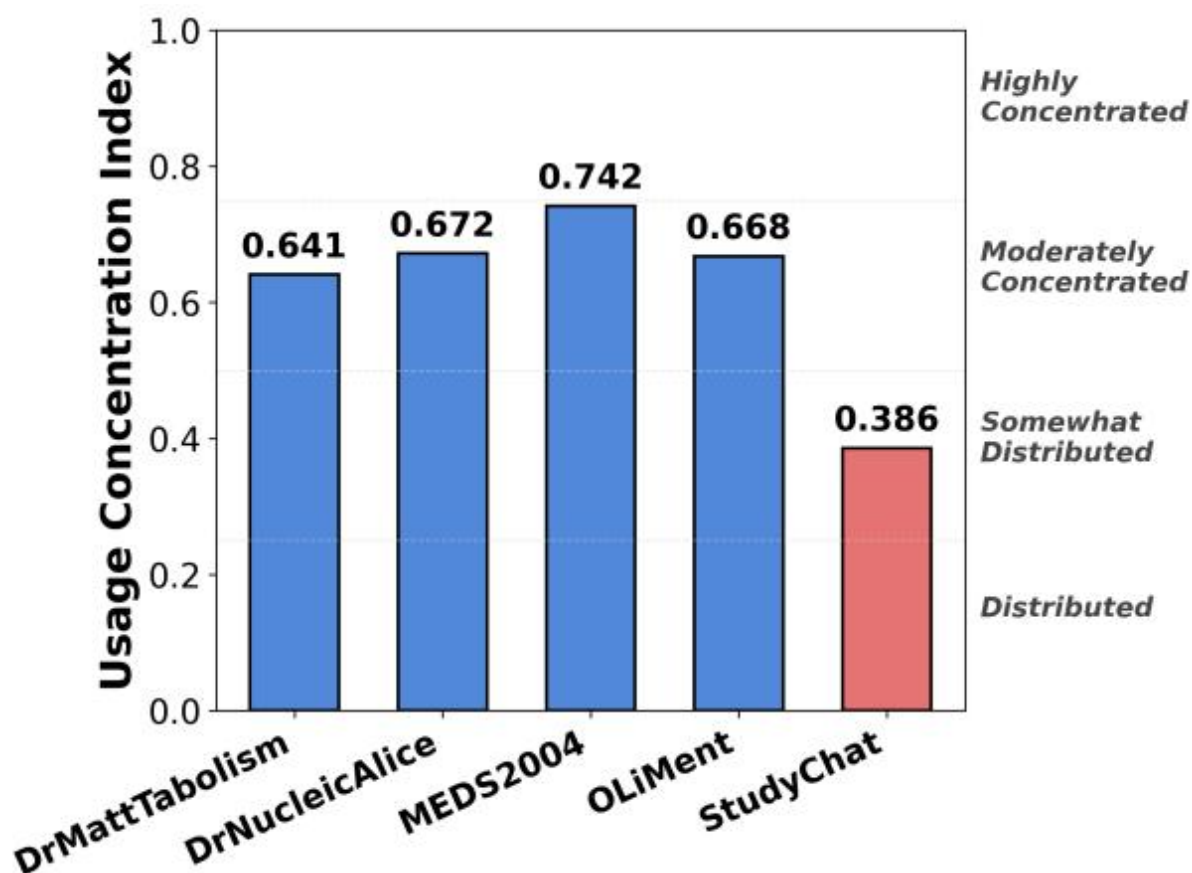


Figure 1: images/19cbad578b5804172562438767b1e874a9af37720d969f25b34f4b879dbab429.jpg

Figure 1: Usage concentration across constrained platforms in blue and the StudyChat dataset in red. Constrained datasets had 0.681 average UCI.

图 1：蓝色表示受限平台的使用集中度，红色表示 StudyChat 数据集。受限数据集的平均 UCI 为 0.681。

tural inevitability when tools are positioned as optional rather than integrated.

工具被定位为可选而非整合时的自然不可避免性。

Unrestricted AI Platform The StudyChat dataset demonstrated UCI of 0.386, representing 43% lower concentration than the constrained platform average. No significant usage spikes were detected throughout the semester, with usage distributed across weeks. However, as other metrics reveal, this temporal distribution did not correspond to improved learning orientation. For student usage across the subject semesters see Appendix 3.

无限制 AI 平台研究聊天数据集展示了 0.386 的 UCI，表示其浓度比受限平台的平均值低 43%。整个学期中未检测到显著的使用高峰，使用情况分布在各个周内。然而，正如其他指标所显示的那样，这种时间分布并未对应于学习导向的改善。有关学生使用情况的详细信息，请参见附录 3。

6 Discussion

6 讨论

6.1 The Detection Gap: Computational Limitations in Academic Integrity

6.1 检测鸿沟：学术诚信中的计算限制

Our ADR metric reveals a fundamental challenge for educational NLP systems: automated detection of assignment-driven usage has issues with both false-positives and false-negatives. Human evaluators readily identify copy-pasted content that LLMs miss by accounting for contextual cues and formatting patterns, understanding when a student's phrasing mirrors assignment language. In contrast, LLMs systematically overestimate assignment dependency, treating legitimate problem-solving discussions as potential homework copying. These false positives may render automated systems practically useless for academic integrity monitoring. These errors are particularly concerning given widespread scaffolding resistance: students who reject pedagogical guidance while extracting homework answers create a false picture of engagement that systems report as active learning dialogue. These findings highlight the need for improved NLP approaches to detecting pedagogically misaligned behaviour in educational dialogue systems.

我们的 ADR 指标揭示了教育自然语言处理系统面临的一个根本性挑战：自动检测以作业为导向的使用存在误报和漏报的问题。人类评估者能够通过考虑上下文线索和格式模式，识别出大型语言模型（LLMs）可能遗漏的复制粘贴内容，并理解当学生的表达方式与作业语言相似时的情况。相比之下，大型语言模型系统地高估了作业依赖性，将合法的问题解决讨论视为潜在的作业抄袭行为。这些误报可能会使自动系统在学术诚信监控方面变得几乎无用。鉴于广泛存在的支架抵抗现象，这些错误尤其令人担忧：那些拒绝教育指导而提取作业答案的学生，会创造出系统报告为积极学习对话的虚假参与表象。这些发现突显了改进自然语言处理方法以检测教育对话系统中与教学不一致行为的必要性。

6.2 Aligning Metrics with Pedagogical Intent

6.2 将指标与教学意图对齐

Our metrics framework enables nuanced interpretation across pedagogically distinct tools. MEDS2004's low learning orientation (0.13) initially appears concerning, yet reflects its intended self-testing function rather than exploratory learning failure. Students requesting practice questions demonstrate active self-assessment, behaviour that educators deploying quiz tools would interpret positively. Our framework allows practitioners to select which metrics matter for their specific context: quiz tools should prioritise engagement over exploratory dialogue, while Socratic tutors might weight scaffolding resistance more heavily.

我们的指标框架使得在教学上不同的工具之间能够进行细致的解读。MEDS2004 的低学习取向（0.13）起初看起来令人担忧，但实际上反映了其预期的自我测试功能，而不是探索性学习的失败。学生请求练习题的行为表明他们正在进行积极的自我评估，这种行为是使用测验工具的教育工作者会积极解读的。我们的框架使实践者能够选择对其特定情境而言哪些指标更为重要：测验工具应优先考虑参与度而非探索性对话，而苏格拉底式导师则可能更重视支架阻力的权重。

This flexibility is a feature, not a limitation. High usage concentration might indicate appropriate exam preparation or problematic procrastination, depending on tool design and pedagogical intent. Our metrics provide educators the detail needed to make these distinctions. While our analysis focuses on educational contexts, mechanisms (metacognitive biases, satisfaction-effectiveness inversion, resistance to scaffolding) are not domain-specific. Similar dynamics may emerge in code assistants or writing tools, though this remains an untested question. Adapting to humanities contexts would require recalibrating detection criteria, as essay-based disciplines

may show different answer-extraction patterns such as seeking thesis statements rather than numerical solutions.

这种灵活性是一个功能，而不是限制。高使用集中度可能表明适当的考试准备，也可能表明存在问题的拖延，这取决于工具的设计和教学意图。我们的指标为教育工作者提供了做出这些区分所需的详细信息。尽管我们的分析集中在教育情境上，但机制（元认知偏差、满意度-效果倒置、对支架的抗拒）并不特定于某个领域。类似的动态可能在代码助手或写作工具中出现，尽管这仍然是一个未经验证的问题。适应人文科学情境将需要重新校准检测标准，因为以论文为基础的学科可能会表现出不同的答案提取模式，例如寻求论点陈述而非数值解。

6.3 Context Shapes Outcomes

6.3 上下文塑造结果

The variation in metrics across our datasets reveals that deployment context, system design, and their interaction are the strongest predictors of pedagogical outcomes more than student preference or technological sophistication.

我们的数据集在指标上的差异表明，部署环境、系统设计及其相互作用是预测教学成果的最强因素，而不是学生偏好或技术复杂性。

DrMattTabolism and DrNucleicAlice demonstrate how implementation differences shape behaviour. Despite serving the same students sequentially, DrMattTabolism's flexible approach yielded shorter conversations with lower scaffolding resistance (0.156), while DrNucleicAlice's stricter Socratic enforcement doubled resistance (0.326) despite longer interactions. This increase reflects both pedagogical structure and temporal fatigue: students completing their second AI-assisted module faced end-of-semester pressures, likely increas-

DrMattTabolism 和 DrNucleicAlice 展示了实施差异如何影响行为。尽管两位教师依次为相同的学生授课，但 DrMattTabolism 灵活的方法导致对话时间更短且支架阻力更低（0.156），而 DrNucleicAlice 更严格的苏格拉底式教学则使阻力翻倍（0.326），尽管互动时间更长。这种增加反映了教学结构和时间疲劳的双重影响：完成第二个人工智能辅助模块的学生面临学期末的压力，这很可能会增加

ing resistance to scaffolding when seeking urgent answers.

在寻求紧急答案时对脚手架的抗性。

MEDS2004 shows how examination framing overrides design. Despite appropriate self-quizzing usage, it showed the lowest learning orientation (0.13) and highest concentration (0.74). Its exam-revision positioning with past papers and rigid sequences prevented exploration even during correct usage. Computationally, MEDS2004's 65 percentage point ADR detection gap demonstrates that current NLP methods cannot distinguish legitimate practice from answer-extraction.

MEDS2004 展示了考试框架如何超越设计。尽管适当使用自我测验，它显示出最低的学习取向（0.13）和最高的专注度（0.74）。其通过以往试卷和固定序列进行考试复习的定位，即使在正确使用的情況下也阻碍了探索。从计算角度来看，MEDS2004 的 65 个百分点 ADR 检测差距表明，当前的自然语言处理方法无法区分合法练习和答案提取。

These patterns demonstrate surface metrics mask behavioural differences. High engagement persists while students avoid learning opportunities. The interaction between implementation (prompt strictness, choreography) and deployment (optional/required, exam/learning, timing) determines learning versus answer-extraction, suggesting deployment strategy should be prioritised for educators.

这些模式表明，表面指标掩盖了行为差异。高参与度持续存在，而学生却回避学习机会。实施（提示严格性、流程设计）与部署（可选/必选、考试/学习、时间安排）之间的互动决定了学习与答案提取，这表明应优先考虑部署策略，以帮助教育者。

6.4 The Unrestricted Access Paradox

6.4 无限制访问悖论

Our findings reveal a fundamental paradox in educational AI deployment that challenges core assumptions about accessibility and learning. The inverse relationship between engagement and learning orientation demonstrates that higher engagement does not translate to better pedagogical outcomes. The unrestricted platform achieved superior engagement metrics yet produced substantially worse learning-orientation score, what we term a “satisfaction-effectiveness inversion.”

我们的研究揭示了教育 AI 部署中的一个根本性悖论，这一悖论挑战了关于可及性和学习的核心假设。参与度与学习导向之间的反向关系表明，更高的参与度并不意味着更好的教学效果。无限制的平台实现了更高的参与度指标，却产生了明显更差的学习导向得分，我们称之为“满意度-效果倒置”。

This paradox aligns with established metacognitive research, the “illusion of fluency” predicted exactly what we observed: smooth, frictionless interactions create false confidence while preventing the desirable difficulties necessary for learning Bjork and Bjork (2011). When students can bypass pedagogical scaffolding, learning-oriented behaviour virtually disappears, replaced by efficient answer extraction. External validation confirms this disconnect: applying our metrics to RECIPE4U (Han et al., 2024), a dataset of student-ChatGPT dialogues with per-response satisfaction ratings, revealed no significant correlation between satisfaction and any pedagogical metric (all $|r| < 0.12$, all $p > 0.2$; $n = 100$). LOI showed near-zero correlation with satisfaction ($r = -0.02$, $p = 0.82$), meaning students rated the AI identically regardless of whether they engaged in exploratory learning or answer-seeking. Full metric breakdowns are provided in Appendix A.5.

这种悖论与已有的元认知研究相一致，“流畅的错觉”准确预测了我们所观察到的现象：流畅、无摩擦的交互虽然会营造出虚假的信心，但却阻止了学习所需的 desirable difficulties（即有益的困难）。Bjork 和 Bjork (2011)。当学生能够绕过教学支架时，以学习为导向的行为几乎完全消失，取而代之的是高效地提取答案。外部验证证实了这种脱节：当我们将我们的指标应用于 RECIPE4U (Han 等, 2024)，这是一个包含学生与 ChatGPT 对话数据集，每条回复都有满意度评分的数据集，结果显示满意度与任何教学指标之间都没有显著相关性（所有 $|r| < 0.12$ ，所有 $p > 0.2$ ； $n = 100$ ）。LOI 与满意度的相关性接近于零（ $r = -0.02$ ， $p = 0.82$ ），这意味着无论学生是进行探索性学习还是直接寻求答案，他们对 AI 的评分几乎是一样的。附录 A.5 中提供了完整的指标分解结果。

6.5 Crisis-Driven Usage as Systemic Failure

6.5 危机驱动的使用作为系统性失败

The temporal concentration patterns reveal misalignment between tool design and usage. These platforms function as emergency services rather than learning companions, a pattern persisting across different contexts suggesting structural inevitability rather than individual choice.

时间集中模式揭示了工具设计与使用之间的不匹配。这些平台更像是应急服务而非学习伙伴，这种模式在不同情境中持续存在，表明这是一种结构性的必然性，而非个人选择的结果。

The behavioural shifts during peak periods such as large reductions in message length and conversation depth indicate students abandon exploratory dialogue when meaningful learning assessment occurs. This isn't a failure of time management but a predictable outcome of positioning pedagogical tools as optional rather than integrated.

在高峰期出现的行为变化，如消息长度和对话深度的大幅减少，表明当有意义的学习评估发生时，学生会放弃探索性对话。这并不是时间管理失败，而是将教学工具定位为可选而非整合的可预见结果。

The contrast with integrated deployment demonstrates that context determines usage patterns

more than technology. Optional tools become crisis management systems, while integration redistributes problematic usage throughout the semester. Neither approach addresses the fundamental misalignment between student goals and educational objectives. Future systems might combine integration with adaptive scaffolding that adjusts pedagogical friction based on real-time behavioural signals from metrics like those we propose.

与集成部署的对比表明，情境比技术更能决定使用模式。可选工具会变成危机管理系统，而集成则会将有问题的使用模式分散在整个学期中。这两种方法都没有解决学生目标与教育目标之间根本性的不匹配问题。未来系统可能会将集成与适应性支架结合，根据我们提出的指标等实时行为信号调整教学摩擦。

7 Conclusion

7 结论

This paper introduces metrics for measuring student behaviour in educational dialogue. These address a critical gap: the absence of tools for evaluating whether systems achieve their intended pedagogical function, not just surface performance. Our results demonstrate that deployment strategy is the strongest predictor of usage patterns, outweighing system design or student preference as a determinant of pedagogical outcomes. Analysis of 500 conversations reveals that students demonstrate high engagement while systematically avoiding learning opportunities, a pattern invisible to standard evaluation. By providing granular behavioural analysis, these metrics enable NLP researchers to move beyond satisfaction measures toward evidence-based evaluation of educational AI.

本文介绍了用于衡量教育对话中学生行为的指标。这些指标填补了一个关键空白：目前缺乏评估系统是否实现其预期教学功能的工具，而不仅仅是表面表现。我们的研究表明，部署策略是使用模式最强的预测因素，其对教学结果的影响超过了系统设计或学生偏好的决定作用。对 500 次对话的分析表明，学生在系统性地回避学习机会的同时表现出高度的参与度，这种模式在常规评估中是不可见的。通过提供细致的行为分析，这些指标使自然语言处理研究人员能够超越满意度测量，转向基于证据的教育人工智能评估。

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8 Limitations

8 局限性

Our study faces several constraints affecting reproducibility and generalisability. Only 20% of our data (StudyChat dataset) is publicly available; the remaining 400 conversations from proprietary platforms cannot be released due to ethics board restrictions and student privacy requirements. We provide detailed methodology, prompts, and statistics to support reproducibility within these constraints.

我们的研究面临一些影响可重复性和可推广性的限制。我们只有 20% 的数据（StudyChat 数据集）是公开可用的；其余 400 次对话来自专有平台，由于伦理委员会的限制和学生隐私要求，无法发布。我们在这

些限制下提供了详细的方法、提示和统计数据，以支持可重复性。

We explored other publicly available educational dialogue datasets (e.g., CIMA, MathDial) as additional validation sources, but these use crowdworkers or LLM-simulated students rather than real students in naturalistic course settings, limiting their suitability for validating behaviours driven by authentic academic pressures. The scarcity of such datasets remains a significant barrier to reproducibility in this area, compounded by our reliance on commercial APIs (GPT-4.1-mini, GPT-5) costing \$145, which may become unavailable.

我们还探索了其他公开可用的教育对话数据集（例如 CIMA、MathDial）作为额外的验证来源，但这些数据集使用众包工作者或由大型语言模型模拟的学生，而非在自然课程环境中使用真实学生，这限制了它们在验证由真实学术压力驱动的行为方面的适用性。此类数据集的稀缺性仍然是该领域可重复性的一个重大障碍，再加上我们依赖于商业 API（如 GPT-4.1-mini、GPT-5），其成本为 145 美元，可能会变得不可用。

Our detection methods, while revealing important patterns, require discipline-specific refinement. Future work should develop tailored detection rules for different fields, incorporating common disciplinary language patterns. Fine-tuning LLM prompts with exemplar interactions from tools like MEDS2004, which show actual AI outputs with past paper questions and typical student response formatting, could substantially improve detection accuracy. Additionally, our sample was exclusively STEM-based; humanities and social science contexts may exhibit different answer-extraction patterns requiring distinct detection approaches.

我们的检测方法虽然揭示了重要的模式，但仍需要根据学科进行细化。未来的工作应开发适用于不同领域的定制化检测规则，结合各学科常见的语言模式。通过使用如 MEDS2004 等工具提供的示例交互来微调大型语言模型（LLM）提示，这些工具展示了实际的 AI 输出，包括以往论文问题和典型的学生成绩格式，可以显著提高检测的准确性。此外，我们的样本仅基于 STEM 领域；人文学科和社会科学的情境可能会表现出不同的答案提取模式，需要采用不同的检测方法。

While UCI captures usage inequality through the Gini coefficient, future work could decompose temporal versus user-based concentration to determine whether crisis-driven patterns are universal or concentrated among heavy users.

虽然 UCI 通过基尼系数捕捉了使用不平等性，但未来的工作可以分解时间因素与用户基础的集中度，以确定危机驱动的模式是普遍存在的，还是集中在重度用户中。

The framework presents dual-use risks: while designed to improve educational practices, our metrics could enable inappropriate student monitoring systems. Our findings should not justify removing AI access entirely, as this could disadvantage students who rely on these tools for legitimate support, including those with disabilities or language barriers.

该框架存在双重用途的风险：虽然旨在改进教育实践，但我们的指标可能会被用于不当的学生监控系统。我们的研究结果不应作为完全取消 AI 访问权限的理由，因为这可能会使依赖这些工具进行合法支持的学生处于不利地位，包括有残疾或语言障碍的学生。

Our analysis does not examine demographic differences in usage patterns. The metrics might inadvertently disadvantage students with learning disabilities, non-native speakers, or those with external responsibilities. We acknowledge that observed answer-seeking behaviours may reflect rational responses to systemic pressures rather than individual failings, and we lack direct student perspectives on their motivations and constraints.

我们的分析并未考察使用模式中的人口统计学差异。这些指标可能会无意中对有学习障碍的学生、非母语者或有外部责任的人造成不利影响。我们承认，观察到的寻求帮助行为可能反映了对系统压力的理性反应，而非个人的失败，而且我们缺乏对学生动机和限制的直接见解。

Our human validation of computational metrics relied solely on the authors' annotations, reflecting a narrow perspective grounded in our own pedagogical traditions. We did not engage external annotators, limiting the cultural and educational diversity of perspectives on what constitutes pedagogical misalignment. Future work should validate these metrics across diverse educational contexts and cultural perspectives on learning. Longitudinal validation would re-

quire objective learning measures, as metacognitive biases resist self-correction without explicit feedback (Hacker et al., 2000; Kruger and Dunning, 1999).

我们的对计算指标的人类验证仅依赖于作者的标注，反映出一种基于我们自身教学传统的狭隘视角。我们没有使用外部标注者，限制了对教学不一致现象的视角的文化和教育多样性。未来的工作应在一个多样化的教育背景和学习文化视角下验证这些指标。纵向验证需要客观的学习测量方法，因为元认知偏见在没有明确反馈的情况下会阻碍自我纠正 (Hacker 等, 2000; Kruger 和 Dunning, 1999)。

Despite these limitations, our work provides critical insights into the gap between pedagogical intentions and actual usage, establishing a foundation for more effective educational AI deployment that acknowledges student realities.

尽管存在这些限制，我们的工作为教学意图与实际使用之间的差距提供了关键的见解，为更有效的教育人工智能部署奠定了基础，该部署承认学生的实际情况。

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